

Sexual selection in plants

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Introduction and core concept

I have been developing an *ovule-gated* model of pollen competition to explain how post-pollination prezygotic reproductive isolation (RI) evolves in flowering plants. My central proposition is that RI does not merely accumulate as a passive, clock-like function of genetic distance. Instead, prezygotic RI can emerge rapidly as a state-dependent “passenger” of the reproductive arena: a by-product of contemporary selection on pollen performance and maternal filtering, rather than the direct target of speciation. The aim of this note is to make that idea precise enough to test. I build a mathematical, simple model, to then derive the main model’s empirical signature—the directional asymmetry of reciprocal crosses, A_{ij} —in the index we measure. I place consider an alternative explanation (unilateral incompatibility) inside the same equations so that the two can be told apart.

The mechanistic model

I consider as the crux of the model a competition ratio defined as:

$$K = \frac{P_{\text{eff}}}{O}, \quad (1)$$

where P_{eff} is the *effective* pollen load—the grains that arrive in time, germinate, and plausibly compete for the same ovules—and O is the number of ovules per ovary. When $K \leq 1$, pollen is at most matched to ovules and competition among grains is weak; when $K > 1$, more competent grains arrive than there are ovules, and some pollen must lose. That losing margin is the raw material for selection and its consequences on speciation.

Why few ovules intensify competition. Suppose effective pollen load scales sublinearly with ovule number, $P_{\text{eff}} = c O^\alpha$ with $0 < \alpha < 1$. Then

$$K = c O^{\alpha-1}, \quad \frac{\partial \log K}{\partial \log O} = \alpha - 1 < 0. \quad (2)$$

Because the exponent is negative, K *falls* as ovule number rises: few-ovule lineages sit in a higher-competition regime even when many-ovule lineages receive more pollen in absolute terms. The condition $\alpha < 1$ is therefore the model’s falsifiable prediction, not the absolute value of any pollen count. The ovule threshold at which competition switches on ($K = 1$) is not universal but emerges from the ecology, $O^* = (\psi c)^{1/(1-\alpha)}$, where ψ absorbs deposition, germination, and timing (Section).

Selection among pollen grains (a standard result). When $K > 1$, only the fraction $q = O/P_{\text{eff}} = 1/K$ of grains can fertilise an ovule. If pollen competitive performance z is approximately Gaussian with standard deviation σ_z , classical truncation-selection theory gives the selection differential among the winning grains as

$$S_z(K) = \begin{cases} 0, & K \leq 1, \\ \frac{\sigma_z \phi(\Phi^{-1}(1 - 1/K))}{1/K} = \sigma_z K \phi(\Phi^{-1}(1 - 1/K)), & K > 1, \end{cases} \quad (3)$$

where ϕ and Φ^{-1} are the standard normal density and quantile functions. This approach follows the textbook “intensity of selection” $i = \phi(z_q)/q$ applied to the saved fraction $q = 1/K$. It rises monotonically with K , so the opportunity for selection on pollen performance grows as ovules become scarce. A beneficial pollen-expressed allele of effect a then changes in frequency at a rate $\Delta p \approx p(1 - p) \kappa a \beta_z(K)$, with gradient $\beta_z(K) = S_z(K)/\sigma_z^2$ and κ mapping the pollen-phase advantage onto whole-plant allele-frequency change. More competition, more sweeps on reproductive genes.

Why high K favours a stricter pistil. Competition alone sharpens the male side; directional RI needs the female side. Let $F_i \geq 0$ be the strictness of lineage i ’s maternal filter. A strict filter raises offspring quality but costs seed set when pollen is scarce. Writing maternal fitness as a product of a saturating seed-set term, a saturating quality term, and a direct cost of strictness, and maximising over F , yields an evolved optimum $F^*(K)$ with the single property we need:

$$\frac{dF^*}{dK} > 0. \quad (4)$$

The logic is intuitive. When K is low, rejecting pollen risks leaving ovules empty, so selection keeps the pistil permissive. When K is high, surplus pollen makes rejection cheap, so the pistil can afford to be choosy. In other words, high competition allows the evolution of female choosiness.

Directional asymmetry: deriving A_{ij}

The model’s simpler prediction is directional, so it must be derived in the directional quantity we actually record. We use the Sobel–Chen reproductive-isolation index. For the directed cross *mother* i , *father* j , let C_{ii} be i ’s conspecific reproductive success and $H_{i \leftarrow j}$ the heterospecific success when i donates the ovules:

$$R_{i \leftarrow j} = 1 - \frac{2 H_{i \leftarrow j}}{H_{i \leftarrow j} + C_{ii}}. \quad (5)$$

Everything simplifies in the single ratio $h_{i \leftarrow j} \equiv H_{i \leftarrow j}/C_{ii}$, the heterospecific success *relative to the mother’s own baseline*:

$$R_{i \leftarrow j} = \frac{1 - h_{i \leftarrow j}}{1 + h_{i \leftarrow j}}, \quad h \in [0, 1]. \quad (6)$$

This is exact. It reads correctly at the limits: $h = 1$ (hybrid as good as conspecific) gives $R = 0$; $h = 0$ (no hybrid success) gives $R = 1$. Each direction is normalised by its *own* mother’s baseline, so the two directions can differ only through quantities that are themselves directional.

The mechanistic content of h . The ovule-gated model puts one directional quantity into h —the maternal filter acting on pollen–pistil mismatch D_{ij} :

$$h_{i \leftarrow j} = \exp(-F_i D_{ij}). \quad (7)$$

Three assumptions follow. First, **(a)** $D_{ij} = D_{ji}$: mismatch is a symmetric property of the *pair*. Second, **(b)** $D_{ii} = 0$: the filter is mismatch-specific—it rejects foreign pollen but passes the mother’s own type, so conspecific success carries no F -dependence and the normalisation in (6) stays clean. And third, **(c)** F_i is a property of the *mother*, set by her reproductive ecology through (4). Assumption (b) must be carefully considered: if a stricter pistil also taxed conspecific pollen, F_i would partly cancel in h and the asymmetry below would be damped.

Exact asymmetry and the sign theorem. Define the reciprocal asymmetry $A_{ij} \equiv R_{i \leftarrow j} - R_{j \leftarrow i}$. With $g(h) = (1 - h)/(1 + h)$ and the *shared* mismatch D_{ij} ,

$$A_{ij} = g(e^{-F_i D_{ij}}) - g(e^{-F_j D_{ij}}). \quad (8)$$

Because g is strictly decreasing and $D_{ij} > 0$,

$$\boxed{\text{sign}(A_{ij}) = \text{sign}(F_i - F_j)}. \quad (9)$$

This recovers the key idea that *symmetric divergence alone cannot make a cross asymmetric*. If $F_i = F_j$, then $A_{ij} = 0$ for any D_{ij} , however large. Genetic distance builds the barrier; it cannot tip it toward one parent. Only a directional maternal trait can—and in this model that trait is filter strictness, which tracks K , which tracks ovule number.

The regression-ready linearisation. For related pairs D_{ij} is small, so $h \approx 1 - F_i D_{ij}$, and expanding g about $h = 1$ (where $g'(1) = -\frac{1}{2}$) gives $R_{i \leftarrow j} \approx \frac{1}{2} F_i D_{ij}$. Hence

$$A_{ij} \approx \frac{1}{2} D_{ij} (F_i - F_j). \quad (10)$$

The factor $\frac{1}{2}$ is the fingerprint of the Sobel–Chen index; an earlier $1 - C$ formulation drops it. The approximation stays within a few percent out to $D_{ij} \approx 0.3$, which covers the divergence range where asymmetry actually lives (below).

From filter strictness to ovule number. Substituting the chain $F = F^*(K)$, $K = c O^{\alpha-1}$, and linearising the filter in $\log O$, define the (positive) sensitivity

$$\beta_F \equiv -\frac{dF}{d \log O} = F'^*(K) K (1 - \alpha) > 0 \quad (\text{positive precisely when } \alpha < 1). \quad (11)$$

Then $F_i - F_j \approx \beta_F (\log O_j - \log O_i)$. Letting mismatch scale with genetic distance to leading order, $D_{ij} \approx \delta_D G_{ij}$, equation (10) becomes the central testable form:

$$\boxed{A_{ij} \approx \underbrace{\frac{1}{2} \beta_F \delta_D}_{\beta_1} G_{ij} (\log O_j - \log O_i)}. \quad (12)$$

The structure is the payoff. The model does *not* predict that asymmetry tracks the ovule-number difference alone. It predicts an **interaction**: asymmetry needs both a

divergence ($G_{ij} > 0$, to build mismatch) *and* an ovule contrast ($\log O_j \neq \log O_i$, to tilt the filter). Remove either factor and $A_{ij} \rightarrow 0$. Only the product $\beta_1 = \frac{1}{2}\beta_F\delta_D$ is identifiable from comparative data; β_F and δ_D are not separable without pollen-load or mismatch measurements.

Holding the ovule contrast fixed, A_{ij} as a function of divergence is **hump-shaped**. It vanishes as $D \rightarrow 0$ (no barrier in either direction) and again as $D \rightarrow \infty$ (both directions saturate at $R \approx 1$, so their difference collapses), peaking at intermediate divergence. Asymmetry is a mid-divergence phenomenon. Empirically this warns that a flat A -versus- G relationship can mean *no mechanism* or *a sample piled up at the saturated end*; we should bin by divergence before drawing conclusions.

An alternative mechanism: unilateral incompatibility

The directional signature (12) is not the model's alone. The Lewis–Crowe rule of unilateral incompatibility (UI) produces a directional barrier that owes nothing to ovules: in crosses between a self-incompatible (SI) and a self-compatible (SC) species, the cross tends to fail when the SI species is the mother and succeed in the reciprocal. To compare like with like, write UI in the same coordinates. Let ρ_i be the maternal pistil's rejection capacity and π_j the paternal pollen's ability to overcome it:

$$h_{i \leftarrow j}^{\text{UI}} = \exp[-\gamma(\rho_i - \pi_j)_+], \quad (x)_+ = \max(x, 0). \quad (13)$$

When the mother is SI (high ρ_i) and the father SC (low π_j), the bracket is large and R is high; the reciprocal sits at the threshold and $R \approx 0$. Letting rejection capacity track SI status $s \in \{0, 1\}$, the asymmetry reduces to

$$A_{ij}^{\text{UI}} \approx \frac{1}{2}\gamma(s_i - s_j). \quad (14)$$

It is set by the SI-status contrast in the maternal-minus-paternal direction. Unlike the ovule model, it does not require genetic distance to scale it, and it does not see ovule number at all.

Telling them apart. Fit one model with both mechanisms present, using $\Delta \log O_{ij} = \log O_j - \log O_i$ and $\Delta s_{ij} = s_i - s_j$:

$$A_{ij} = \beta_0 + \beta_1[G_{ij} \cdot \Delta \log O_{ij}] + \beta_2 \Delta s_{ij} + \beta_3 \Delta \log O_{ij} + \varepsilon_{ij}. \quad (15)$$

The two mechanisms leave different fingerprints:

	Ovule-gated filter	Unilateral incompatibility
Sets the <i>sign</i> of A_{ij}	ovule contrast $\Delta \log O$	SI-status contrast Δs
Needs genetic distance to scale	yes ($G \times \Delta \log O$)	no
Diagnostic coefficient	$\beta_1 > 0$	$\beta_2 > 0$
Predicted ≈ 0	β_2, β_3	β_1
Vanishes at high divergence	yes (saturation)	no (S-locus persists)
Vanishes when $O_i = O_j$	yes	no

The identifiability catch. If, across the available pairs, the few-ovule species *are* the self-incompatible ones, then $\Delta \log O$ and Δs are collinear and β_1, β_2 cannot be separated; the regression will reward whichever term is entered first. Breaking the tie requires pairs where the two contrasts point in different directions, or external information: the S-locus genotype, or the anatomical site of pollen-tube arrest—a stigma or upper-style block is the UI signature, while deeper, ovule-level failure points to the filter model. Measuring the collinearity between $\Delta \log O$ and Δs in the present data is the first thing to do before trusting any asymmetry result.

The postzygotic contrast

I expect postzygotic barriers to behave differently. Modelling the Dobzhansky–Muller incompatibility load as $B_{ij} \sim \text{Poisson}(\lambda_B T_{ij}^\gamma)$ and $R_{ij}^{\text{post}} = 1 - e^{-hB_{ij}}$, the barrier is essentially symmetric in cross direction and should track divergence time or genetic distance more closely than maternal state. The cleanest version of the whole proposal is therefore a *decoupling*: prezygotic RI tracks maternal K (hence ovule number) and shows the interaction (12), while postzygotic RI tracks G_{ij} and shows little direction. If both barriers track ovule number and maternal direction in the same way, the model needs revision.

Methodological approach and missing data

Mating system is a separate lever on K . The deposition fraction $\psi = \delta\gamma\tau$ in $P_{\text{eff}} = \psi A$ is the channel mating system controls. An obligate selfer receives little outcross pollen, so $\delta \rightarrow 0$ and K collapses for reasons that have nothing to do with ovule number; worse, its self pollen is genetically near-uniform, so the variance that sexual selection needs is absent. Mating system and ovule number are thus separate multiplicative inputs to the same K . To isolate the ovule lever, the analysis should hold the mating-system lever roughly fixed and high—that is, restrict to outcrossers, retaining mixed-maters only with explicit justification and excluding selfers. This is not bookkeeping; it follows from the model’s own definition of K .

The withheld pollen-load data become a calibration problem, not a dependency. Direct pollen-load measurements are unavailable. Rather than let that derail the theory, I treat effective load as a latent quantity and ask whether the predictions survive across biologically plausible ranges. Concretely: vary the deposition fraction ψ over (say) $\{0.001, \dots, 0.10\}$ and the scaling exponent α over $\{0.25, \dots, 1.25\}$, and treat $\log K_i$ as a phylogenetically structured latent variable, $\log K_i = \mu_K + (\alpha - 1) \log O_i + u_i + \epsilon_i$, so that uncertainty in K propagates into the RI prediction. The model earns trust if the few-ovule signal persists across a broad band of ψ rather than at one tuned value, and the result hinges on $\alpha < 1$ rather than on any exact pollen count.

Next steps: simulations

The simulation programme has three layers, each tied to a quantity above. First, an *ecological competition engine* with no RI: draw overdispersed pollen loads, let the top O grains win, and confirm that few-ovule (high- K) lineages experience the stronger selection

of (3) across the ψ - α grid. Second, *evolving maternal filters* under independent, linked, and pleiotropic genetic architectures: let F evolve toward $F^*(K)$, perform reciprocal crosses in silico, and check that the recovered A_{ij} obeys the sign theorem (9) and the interaction (12), with pleiotropy > linkage > independence in the speed of passenger-RI generation. Third, *realistic data damage* on simulated phylogenies: impose sparse reciprocals, summary-only RI, and clade-structured ovule states, then ask under which regimes the discriminating regression (15) recovers $\beta_1 > 0$ with acceptable power and false-positive rate—and under which the ovule and UI signals become inseparable. The aim is to learn, before touching the real data again, exactly which datasets can carry the claim in the title and which can only suggest it.

Note on verification. The algebraic reduction in (6), the sign theorem (9), the accuracy of the linearisation (10), and the hump-shaped dependence of A_{ij} on divergence were checked numerically before being written here. References (Sobel & Chen on the RI index; Lewis & Crowe on unilateral incompatibility; standard truncation-selection theory) are cited in prose and remain to be formalised.